Patient-robust discovery of combinatorial cell-surface targets in prostate cancer

AND-gate and NOT-gate marker pairs from single-cell data, benchmarked against PSMA-PSCA

William Guesdon

2026-07-13 16:46

Abstract Single-antigen therapies against solid tumors are limited by on-target off-tumor toxicity, because few surface antigens are truly absent from all healthy tissue. Combinatorial (logic-gated) targeting requires two conditions before a cell is engaged, which can recover specificity. We scan a curated cell-surface panel in single-cell data for pairs that discriminate prostate cancer cells from healthy human cell types, under AND (both markers on the tumor) and NOT (an activator spared by a blocker on healthy cells) logic. Pairs are scored per patient and summarized across patients, so no high cell-count patient drives a result. The preclinically validated PSMA-PSCA balanced-signalling pair is recovered as a positive control: each marker alone carries a large extra-prostatic transcript liability, and the AND gate collapses it. We report a Pareto frontier of surface-accessible pairs with higher per-patient malignant co-detection than PSMA-PSCA at comparable predicted normal-tissue liability, with Human Protein Atlas evidence supporting membrane-localization plausibility. These are transcript-level, hypothesis-generating rankings, not evidence of therapeutic safety.

Table of contents

1 Background

A major challenge for antigen-directed therapies in solid tumors is achieving tumor coverage without clinically important on-target, off-tumor activity. A single tumor antigen is rarely absent from every healthy tissue, so single-target CAR-T, T-cell engagers, and antibody-drug conjugates carry this risk. Combinatorial, logic-gated targeting requires two conditions before a cell is engaged and can recover the missing specificity [1]–[3]. Prostate adenocarcinoma is the natural test bed: the PSMA (FOLH1) x PSCA pair is a preclinically validated AND-gate split-signal CAR [1], so a systematic scan of a curated panel can be checked against a known answer, and STEAP1 anchors a second axis with a STEAP1 x CD3 T-cell engager (xaluritamig) in trials [4]. That trial establishes STEAP1 as druggable in prostate cancer; it does not validate a PSMA x STEAP1 gate.

Computational logic-gated antigen discovery is not itself new: pan-cancer single-cell pair scans [5] and a breast-cancer AND/OR/NOT circuit search [6] precede this work. Our contribution is the combination of (i) per-patient scoring rather than pooled cells, (ii) a matched benign-prostate control that tests tumor specificity in the same samples, (iii) an assay-matched, donor-robust healthy reference so normal-tissue liabilities are comparable across platforms and not driven by a single donor, and (iv) recovery of the PSMA-PSCA benchmark before any new pair is nominated.

2 Data

- **Tumor discovery cohort** — 68,322 cells from **24 patients** with localised, hormone therapy-naive prostate adenocarcinoma [7]. Malignant identity uses the authors' copy-number and signature-based annotation: **3,590 malignant cells** (scDblFinder singlets; see Methods), with **16,249 adjacent-benign / normal prostate epithelial cells** retained as a matched same-patient control.
- **Healthy reference** — Tabula Sapiens 2.0 [8], [9], pulled through the CELLxGENE Census (release 2025-01-30) and restricted to the **10x 3' v3** assay to match the tumor cohort chemistry, since a raw-count positivity threshold is not comparable across assays: **1,025,717 cells**, **173 cell types** across **65 tissue labels** (about 30 organs); **442 donor-replicated tissue-by-cell-type populations** (at least 2 donors each) form the safety denominator.
- **Protein evidence** — Human Protein Atlas subcellular localization and normal-tissue RNA [10], used to qualify surface accessibility and cross-check normal-tissue liabilities.
- **Panel** — 29 curated cell-surface markers scanned (see Methods); secreted markers (KLK3/PSA, KLK2, ACPP) are kept only as comparators.

3 Methods

- **Doublet removal (primary input).** scDblFinder, run per patient in a Bioconductor container, flagged **4.8%** of tumor cells as doublets (5.6% of malignant cells); these are removed before scoring, so a two-cells-as-one capture cannot fake AND co-expression (Rule 8). Re-scoring on the

65,040 retained single cells moves the key pairs' AND coverage by at most 1.0 percentage points, so the finding is not a doublet artifact.

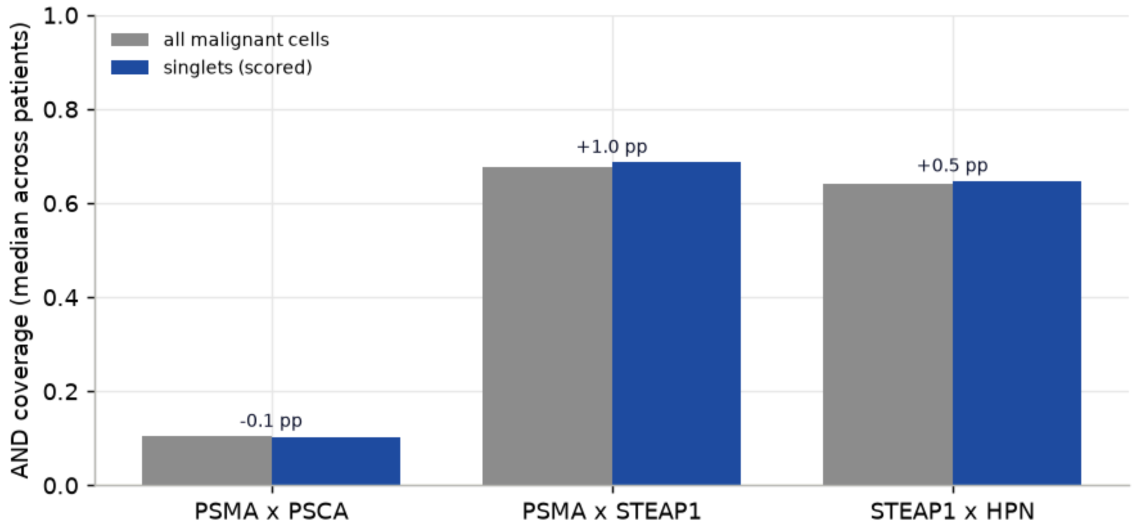

Figure 1: Doublet robustness. Median per-patient AND coverage of the key pairs computed on all malignant cells (grey) versus the scDblFinder singlets actually used for scoring (blue). The bars are near-identical and the change (annotated, percentage points) is at most one point, so the co-expression signal is not an artifact of two cells captured as one; removing doublets nudges the nominated pairs slightly upward.

- **Positivity** is a per-cell raw count of at least 1 (single-cell detection, not a threshold on a pooled or averaged expression value); the threshold is swept for sensitivity (below).
- **Per-patient scoring (never pooled).** For markers A, B and each patient's malignant cells, AND coverage is `P(A+ and B+ | malignant)` computed within that patient, then summarized across patients by the lower decile (Q0.10, the robustness floor) and the median. A patient with thousands of cells cannot dominate a pair's score.
- **Three references per pair.** (1) malignant coverage; (2) **matched benign prostate** from the same cohort, so a pair expressed across all prostate cells is penalised, not just one specific to the tumor;

  (3) **healthy liability** as the worst-case co-positive fraction over Tabula Sapiens populations. The healthy side is assay-matched to the tumor (10x 3' v3 only) and made **donor-robust**: within each tissue-by-cell-type population the fraction is computed per donor and summarized by the median across donors, keeping only populations with at least two donors, so no single donor or tiny population sets a worst case. Every worst-liability call carries its supporting donor and cell counts. Liabilities are reported both across all organs and **excluding the prostate** (the target organ is dispensable; every other organ is the dose-limiting risk).
- **Gate algebra.** For each group a single matrix product over the panel positivity matrix yields AND, NOT (`A+ and B-`) and OR counts for every pair at once. These Boolean gates are screening abstractions of candidate receptor logic; they do not model antigen-density thresholds, residual

single-arm activity, or the architecture-specific signalling of a balanced split-signal CAR, synNotch, or Tmod circuit.

- **Ranking.** Pairs are placed on a Pareto frontier of per-patient coverage floor (higher better) against worst extra-prostatic liability (lower better). Only surface-accessible markers are nominatable.
- **Reproducibility.** Every number below is read from a committed table in `results/tables/`; the report redraws its figures from those tables with no network or model call.

4 Results

4.1 Single surface antigens each carry a large normal-tissue liability

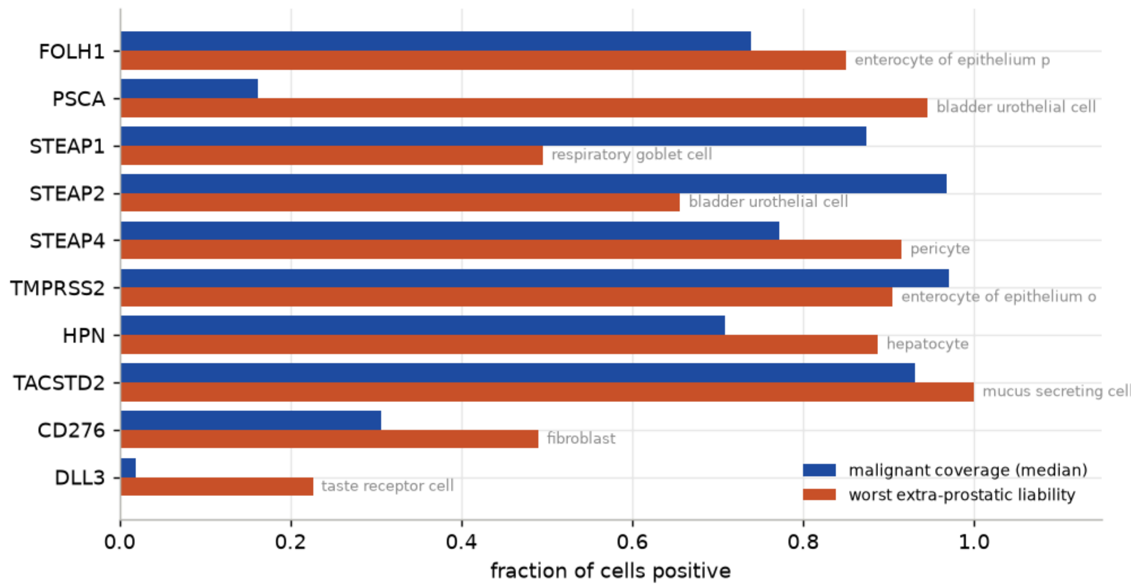

Figure 2: Single surface antigens: per-patient malignant coverage (median) against the worst-case extra-prostatic normal-tissue co-detection. Every strong single antigen also carries a large normal-tissue transcript liability; the recovered worst tissues are concordant with known biology (FOLH1/PSMA to duodenum, PSCA to bladder urothelium).

- Each strong single antigen is broadly positive on the tumor but also on at least one healthy cell type: the scan independently recovers PSMA's duodenal and PSCA's bladder-urothelial expression, a qualitative sanity check that the recovered tissues match known biology, not a claim that the fractions are quantitatively exact.

4.2 The AND gate recovers PSMA-PSCA and collapses its single-agent liabilities

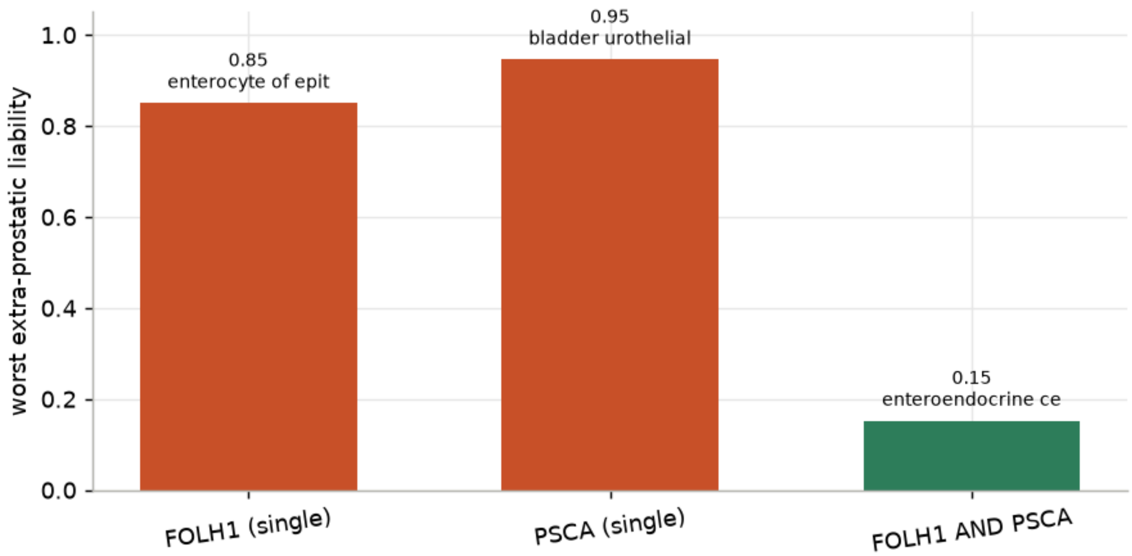

Figure 3: Positive control. PSMA (FOLH1) and PSCA are each broadly detected off the prostate (duodenum, bladder), but requiring both at once (AND) collapses the worst extra-prostatic co-detection roughly sixfold, recapitulating the direction of the preclinical rationale for the split-signal PSMA x PSCA CAR.

- **Recovered.** PSMA alone reaches **0.85** co-detection in duodenal enterocytes and PSCA alone **0.95** in bladder urothelium; the **PSMA and PSCA** gate drops the worst extra-prostatic co-detection to **0.15** (a reduction of 0.79), the expected direction of the preclinical benchmark.
- **Against a random baseline.** 57% of the sampled surface pairs (drawn from the same curated panel) score worse than PSMA-PSCA on extra-prostatic selectivity, defined as median malignant co-detection minus worst extra-prostatic liability. This is a weak signal: a pair can also look clean by carrying little tumor coverage, so it is a sanity check, not proof of specificity.
- Its low malignant co-detection (0.10 median) is consistent with limited PSCA transcript detection in this dataset; protein-level measurement would be needed to separate technical non-detection from biological heterogeneity. It motivates looking for higher-coverage pairs of comparable liability.

4.3 A Pareto frontier of surface pairs with higher coverage than PSMA-PSCA

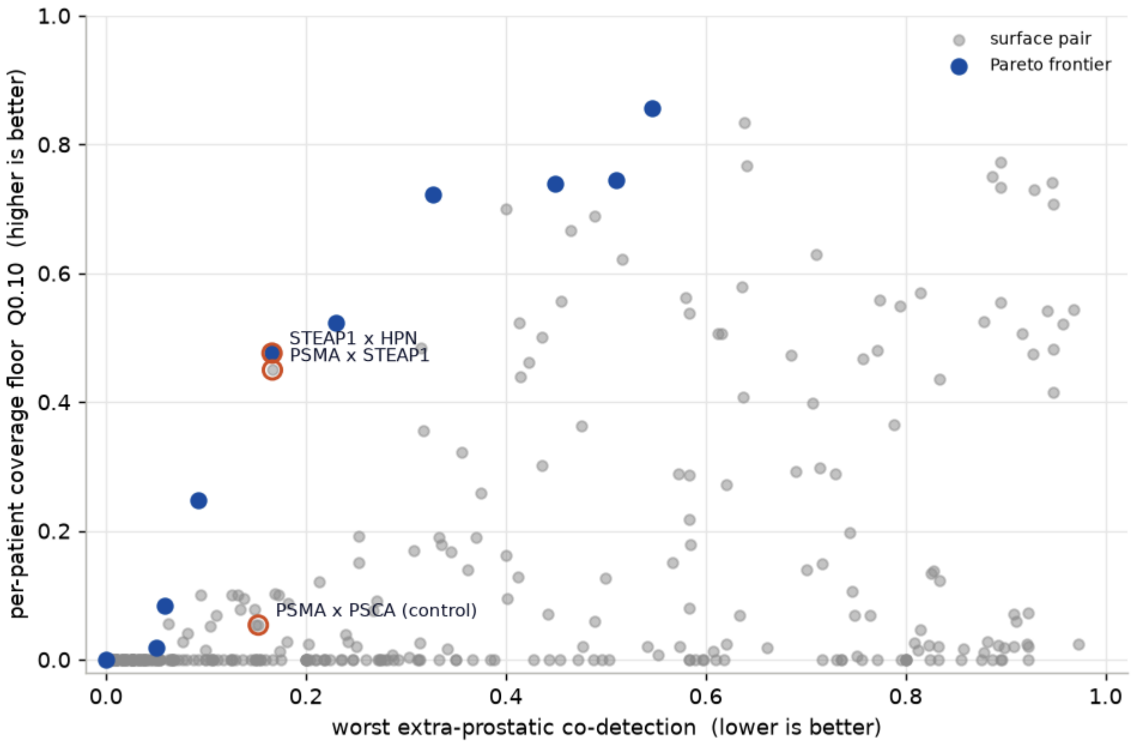

Figure 4: Surface-accessible AND pairs: per-patient coverage floor (Q0.10) against worst extra-prostatic liability. Upper-left is better. Frontier pairs are filled; the PSMA-PSCA positive control and the two co-lead nominations are labelled. Secreted markers and intracellular prostein are excluded as non-targetable.

Two surface-accessible candidates show favourable transcriptomic coverage–liability trade-offs relative to the control: higher malignant co-detection at a comparable predicted maximum normal-tissue liability.

- **PSMA x STEAP1 — translationally tractable candidate.** Both antigens have clinical-grade binders. Per-patient coverage floor **Q0.10** = **0.45**, median **0.69** (**6.7x** the PSMA-PSCA median). Its worst extra-prostatic co-detection is **0.17** in mucus secreting cell (5 donors), versus 0.85 / 0.50 for the single antigens and 0.15 for PSMA-PSCA. Whether that residual liability is clinically acceptable cannot be judged from transcript data.
- **STEAP1 x HPN — Pareto-front candidate.** Coverage floor **Q0.10** = **0.48**, median **0.65**, worst extra-prostatic co-detection **0.17** in hepatocyte — the lowest maximum extra-prostatic co-detection among the high-coverage pairs analysed. HPN is not tumor-restricted (elevated in liver, kidney, pancreas), so this pair needs protein-level validation in those tissues in particular.
- **Higher in malignant than adjacent-benign prostate.** Both candidates show more co-detection in annotated malignant than in matched benign prostate epithelium (malignant-minus-benign 0.60 for PSMA x STEAP1).

4.4 Robustness: per patient and across positivity thresholds

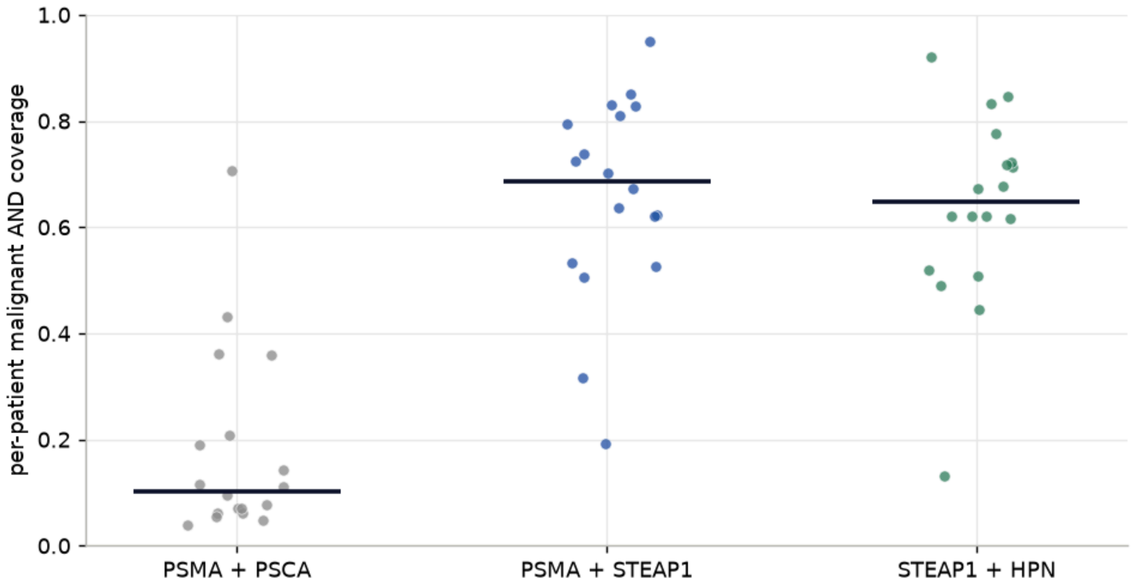

Figure 5: Per-patient AND co-detection of the malignant compartment for the positive control and the two candidates. Each point is one patient; the bar is the median. PSMA x STEAP1 and STEAP1 x HPN hold a per-patient coverage floor (Q0.10) above 0.44 across scored patients, while PSMA x PSCA is uniformly low (limited PSCA transcript detection).

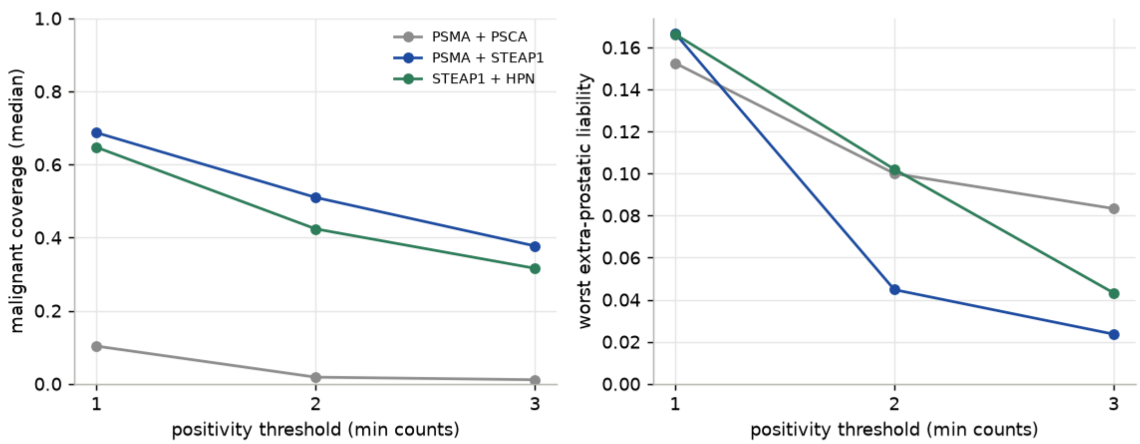

Figure 6: Positivity-threshold sensitivity. As the per-cell call is tightened from >=1 to >=3 counts, malignant co-detection falls modestly while the worst extra-prostatic co-detection falls too: the candidates stay well covered and their normal-tissue liability drops, whereas PSMA x PSCA loses coverage. Positivity is a per-cell detection threshold, swept here, not a pooled-mean call.

- Both candidates survive threshold tightening: malignant co-detection decreases gradually while the worst extra-prostatic co-detection falls, so the selectivity ranking is not an artifact of the loosest positivity call.

Residual off-tumor liabilities of PSMA x STEAP1 are named, not hidden. The highest extra-prostatic healthy populations (donor-robust AND co-detection, with donor and cell support):

tissue	cell_type	coverage	n_donors	n_cells
respiratory system	mucus secreting cell	0.167	5	269
small intestine	paneth cell of epithelium of small intestine	0.143	5	1657
tongue	salivary gland cell	0.14	2	734
small intestine	intestinal crypt stem cell of small intestine	0.095	5	531
lung	respiratory goblet cell	0.071	4	978
respiratory system	basal cell	0.065	5	10003

4.5 NOT gates: a negative result

No NOT gate (activator on the tumor, blocker sparing a normal tissue) achieved both high malignant coverage and low predicted normal-tissue escape. The highest-ranking exploratory combinations still retain a large activator-positive / blocker-negative fraction in normal tissue (worst extra-prostatic co-detection shown), so none is an attractive candidate; the blocker-negative call also depends on trusting scRNA dropout. This is reported as a negative result:

activator	blocker	tumor_q10	tumor_median	worst_healthy_xprostate	worst_xp_celltype
STEAP1	SLC34A1	0.739	0.874	0.496	respiratory goblet cell
STEAP1	CDH16	0.739	0.874	0.496	respiratory goblet cell
STEAP2	MUC1	0.6	0.843	0.477	fibroblast
STEAP1	MUC1	0.543	0.732	0.411	fibroblast
STEAP2	SLC34A1	0.936	0.968	0.653	bladder urothelial cell

5 Limitations

- **Transcript co-detection is a selectivity surrogate, not therapeutic safety.** scRNA measures mRNA, not surface antigen density, extracellular epitope availability, binding affinity, shedding, internalization, the minimum antigen density for activation, or residual single-arm signalling. HPA localization supports membrane-localization plausibility but does not qualify targetability; same-cell surface co-expression needs CITE-seq, multiplex IHC, or dual-color flow to confirm.
- **Statistical uncertainty is not yet quantified.** Per-patient coverage is summarized by the median and Q0.10 without confidence intervals, and the healthy worst case is a maximum over

hundreds of populations. Per-patient beta-binomial intervals, bootstrap-over-patients intervals, and a donor-level upper confidence bound for liability are the planned next step; the current rankings are point estimates.

- **Assay-matched but single-assay.** The healthy reference is restricted to 10x 3' v3 to match the tumor, so Smart-seq2 and other-chemistry cells in Tabula Sapiens are excluded and a liability seen only on another platform would be missed. Dissociation also selectively loses fragile populations.
- **NOT-gate calls trust B-negatives**, which are dropout-sensitive, and none cleared the liability bar (a negative result).
- **One cohort, localised disease.** The discovery cohort is hormone-naive localised prostate cancer; replication in an independent and a castration-resistant / neuroendocrine cohort is future work.
- **Malignant labels are the authors' CNV/signature calls**, adopted rather than re-derived here.
- **The PSMA-PSCA benchmark is preclinical.** It is a safety/selectivity benchmark, not a coverage benchmark; its low co-detection is consistent with limited PSCA transcript detection in this dataset.

6 Conclusion

This analysis is a patient-aware, transcriptome-based prioritization of candidate combinatorial antigens in localized prostate cancer. The PSMA-PSCA benchmark shows the expected collapse in normal-tissue transcript co-detection relative to its single markers, recovered blind. PSMA x STEAP1 and STEAP1 x HPN show higher malignant co-detection than the benchmark within the discovery cohort, at differing predicted normal-tissue liabilities. These are hypothesis-generating rankings, not evidence of therapeutic safety. Independent tumor cohorts, an assay-matched normal reference at protein level, and same-cell measurements (CITE-seq, multiplex IHC) are required before either pair can be called a validated target.

7 Reproducibility

The pipeline runs from a clean clone with `uv`; every table in `results/tables/` is regenerated by the numbered scripts in `scripts/`, and this report redraws its figures from those tables. See the README for the exact command sequence.

8 References

Bibliography

[1] C. C. Kloss, M. Condomines, M. Cartellieri, M. Bachmann, and M. Sadelain, "Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells," *Nature Biotechnology*, vol. 31, no. 1, pp. 71–75, 2013, doi: 10.1038/nbt.2459.

[2] K. T. Roybal, L. J. Rupp, L. Morsut, and others, "Precision tumor recognition by T cells with combinatorial antigen-sensing circuits," *Cell*, vol. 164, no. 4, pp. 770–779, 2016, doi: 10.1016/j.cell.2016.01.011.

[3] J. H. Cho, J. J. Collins, and W. W. Wong, "Universal chimeric antigen receptors for multiplexed and logical control of T cell responses," *Cell*, vol. 173, no. 6, pp. 1426–1438, 2018, doi: 10.1016/j.cell.2018.03.038.

[4] W. K. Kelly, D. C. Danila, C.-C. Lin, and others, "Xaluritamig, a STEAP1 x CD3 XmAb 2+1 immune therapy for metastatic castration-resistant prostate cancer: results from dose exploration in a first-in-human study," *Cancer Discovery*, vol. 14, no. 1, pp. 76–89, 2024, doi: 10.1158/2159-8290.CD-23-0964.

[5] J. Kwon, J. Kang, and others, "Single-cell mapping of combinatorial target antigens for CAR switches using logic gates," *Nature Biotechnology*, vol. 41, pp. 1593–1605, 2023, doi: 10.1038/s41587-023-01686-y.

[6] S. Madan, T.-G. Chang, E. Ruppin, and others, "Single-cell-guided identification of logic-gated antigen combinations for designing effective and safe CAR therapy," *bioRxiv*, 2025, doi: 10.1101/2025.03.19.644074.

[7] E. Apostolov *et al.*, "Single-Cell and Spatial Transcriptomic Profiling Reveals Epithelial Functional States and Fibroblast Phenotypes in Hormone Therapy–Naïve Localized Prostate Cancer," *Cancer Research*, vol. 86, no. 8, pp. 1836–1853, 2026, doi: 10.1158/0008-5472.CAN-25-1202.

[8] The Tabula Sapiens Consortium and S. R. Quake, "Tabula Sapiens reveals transcription factor expression, senescence effects, and sex-specific features in cell types from 28 human organs and tissues," *bioRxiv*, 2024, doi: 10.1101/2024.12.03.626516.

[9] The Tabula Sapiens Consortium, "The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans," *Science*, vol. 376, no. 6594, p. eabl4896, 2022, doi: 10.1126/science.abl4896.

[10] M. Uhlén, L. Fagerberg, B. M. Hallström, and others, "Tissue-based map of the human proteome," *Science*, vol. 347, no. 6220, p. 1260419–1260420, 2015, doi: 10.1126/science.1260419.